

Programmable Interval Timer (8253/8254)

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Course Title: Peripherals, Interfacing and Embedded Systems

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Lecture References:

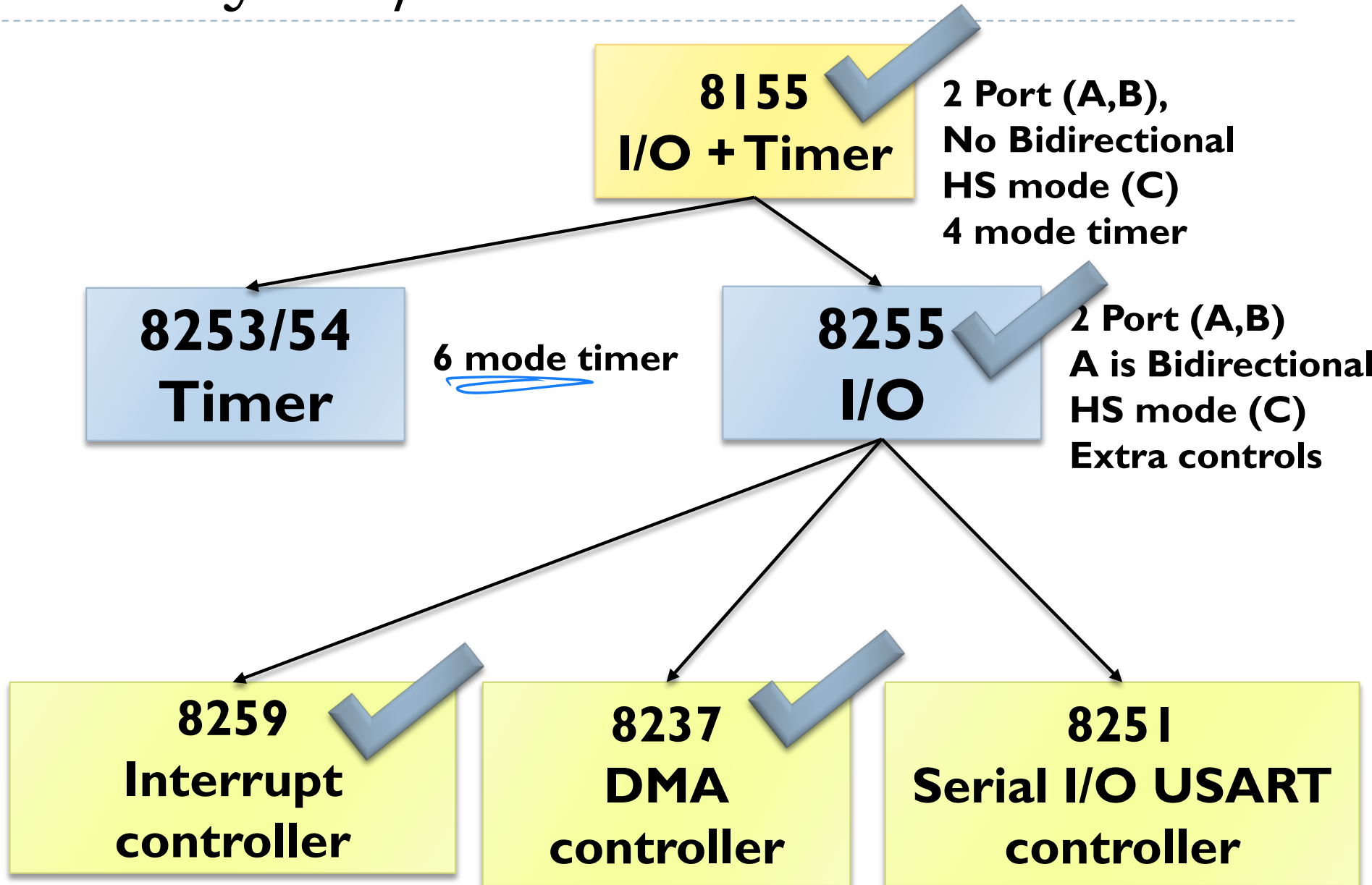
- ▶ **Book:**

- ▶ *Microprocessors and Interfacing: Programming and Hardware,*
Author: Douglas V. Hall

- ▶ **Lecture Materials:**

- ▶ *K.Vijay Kumar, Assistant Professor, Dept. of ECE*

Hierarchy of I/O Control Devices



Why Programmable Interval Timer

Not Possible To Generate Accurate Time Delays Using Delay Routines in 8086

that's why timing func has to be produced by the user itself



that's why we use external timer devices

defined by the user

Intel's Programmable Counter/ Timer Device (8253/8254) Facilitates

- **Accurate Time Delays**
- **Minimizes Load On Mp**
- **Real Time Clock**
- **Event Counter**
- **Digital One Shot**
- **Square Wave Generator**
- **Complex Waveform Generator**

8253 vs. 8254 PIT

8253

- 8253 can operate at frequency from dc to 2mhz

can only read value from one counter at a time

8254-ADVANCED VERSION OF 8253

- 8254 can operate with higher clock Frequency Range (DC To 8 Mhz AND 10 Mhz FOR 8254-2)
- Includes Status Read Back Command That Latches The Count And Status Of Counters

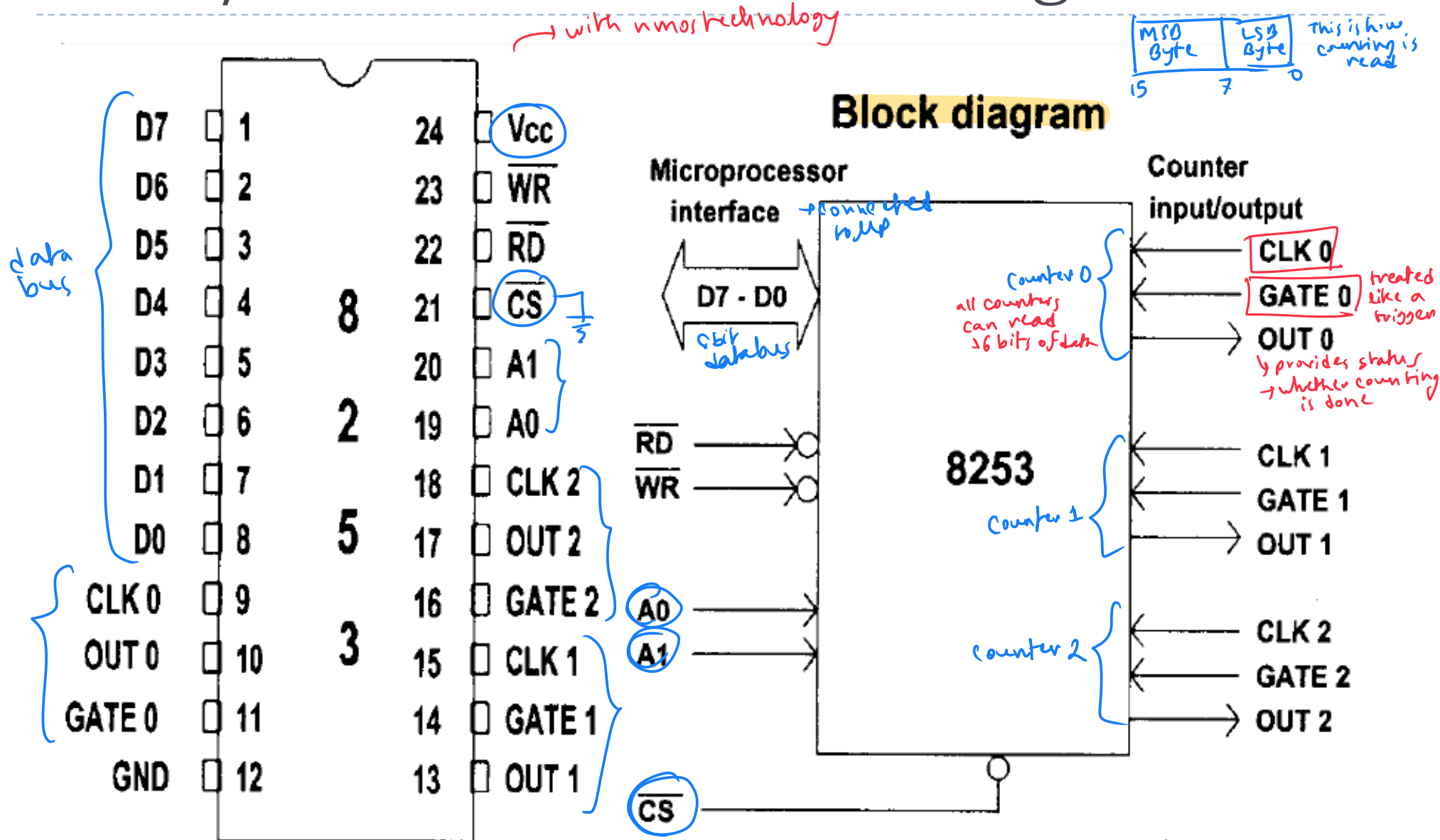
can read values from three counters simultaneously

8253 vs. 8254 PIT

- ▶ Both compatible with All Intel and Most other Microprocessors and Handles Inputs up to 10 MHz
- ▶ The Intel 8254 is a counter/timer device designed to solve the common timing control problems in microcomputer system design. *how many instructions were executed, how many bytes were counted, how many bits have been read from I/O device, how much time is required to process one byte of data*
- ▶ It provides Three Independent 16-bit counters, each capable of handling clock inputs up to 10 MHz *can count up to 2¹⁶ operations*
- ▶ Binary or BCD counting
- ▶ Single a +5V Supply and Standard Temperature Range
- ▶ Six Programmable Counter Modes and All modes are software programmable. The 8254 is a superset of the 8253.
- ▶ Used for controlling real-time events such as real-time clock, events counter, and motor speed and data direction control.

software dependent

8253/8254 Pin and Block Diagram



8253/8254 Pin and Block Diagram

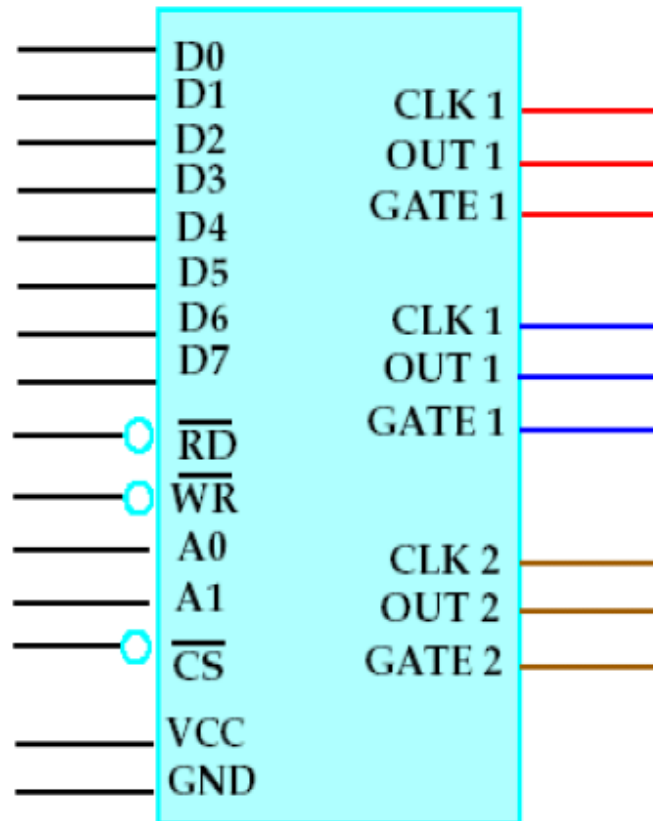
A_1	A_0	
0	0	→ Counter 0
0	1	→ Counter 1
1	0	→ Counter 2
1	1	→ Contr. 1 register

with hmos technology



Pin-out

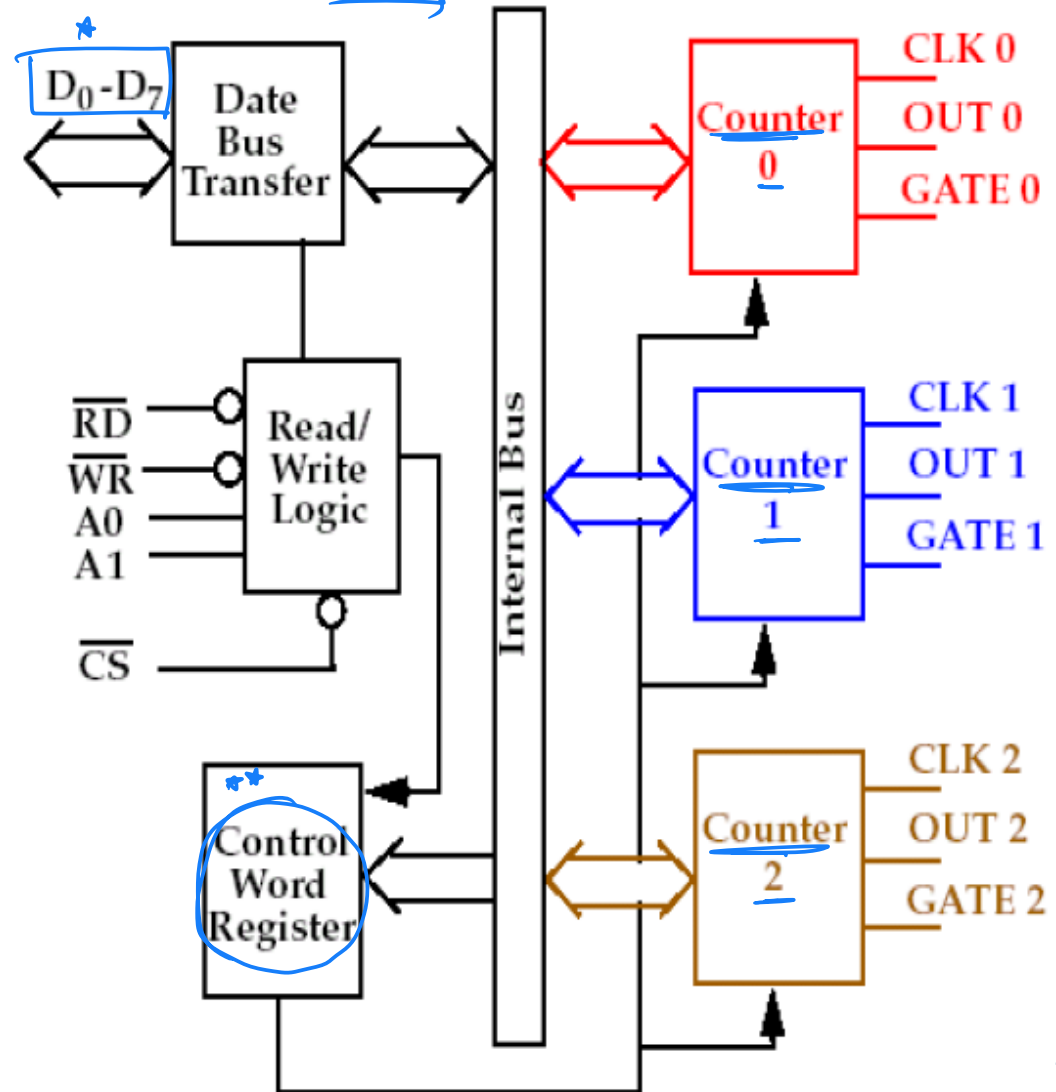
no difference with 8253



how much count has elapsed

Internal structure

counting instruction comes



8253/8254 Pin and Block Diagram

▶ **Data Buffers:**

- ▶ 8 Bit Bidirectional D0-D7 Connected To Data Bus of Microprocessor
- ▶ In → Reads Data from Peripheral
- ▶ Out → Writes Data to Peripheral

▶ **Control Word Register:**

- ▶ Accepts 8 Bit Control Word Written by Microprocessor
- ▶ Can Only Be Written (**Read Not Possible !!**)
- ▶ Control Word Chooses One of the Six Modes of Operation

8253/8254 Pin and Block Diagram

► Counters:

- Three Counters – C1, C2 & C3
- Each 16 Bit Identical Presetable
- Down Counter Operates In BCD/Hex
- Controlled By Loading Count To Command Word Register
- “On The Fly” Reading is possible *count command*
readback command

► Control Logic:

- $\overline{CS}$ – Logic 0 – Enables 8254
- $\overline{RD}$ – Logic 0 – Tells Microprocessor Reads Count from 8254
- $\overline{WR}$ – Logic 0 – Tells Microprocessor Writes Count/ Command into 8253/8254
- AI, A0 – Address Input Pins To Select Modes and Counters

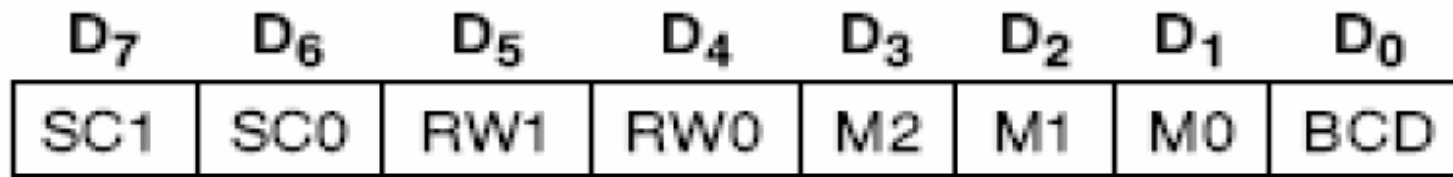
These 5 pins receive data from μP

8253/8254 Operation

$\overline{CS}$	$\overline{RD}$	$\overline{WR}$	A1	A0	OPERATION
0	1	0	0	0	Write Counter 0
0	1	0	0	1	Write Counter 1
0	1	0	1	0	Write Counter 2
0	1	0	1	1	Write Control Word
0	0	1	0	0	Read Counter 0
0	0	1	0	1	Read Counter 1
0	0	1	1	0	Read Counter 2
0	0	1	1	1	No Operation (Tristated)
0	1	1	X	X	No Operation (Tristated)
1	X	X	X	X	<u>8254 Not Selected</u>

Control Word Format

→ this will be stored in control word register.



SC—Select Counter

SC1	SC0	
0	0	Select Counter 0
0	1	Select Counter 1
1	0	Select Counter 2
1	1	<u>Read-Back Command</u> (see Read Operations)

M—Mode

M2	M1	M0	
0	0	0	Mode 0
0	0	1	Mode 1
X	1	0	Mode 2
X	1	1	Mode 3
1	0	0	Mode 4
1	0	1	Mode 5

RW—Read/Write

RW1	RW0	
0	0	<u>Counter Latch Command</u> (see Read Operations)
0	1	Read/Write <u>least significant byte only</u>
1	0	Read/Write <u>most significant byte only</u>
1	1	Read/Write <u>least significant byte first, then most significant byte</u>

BCD

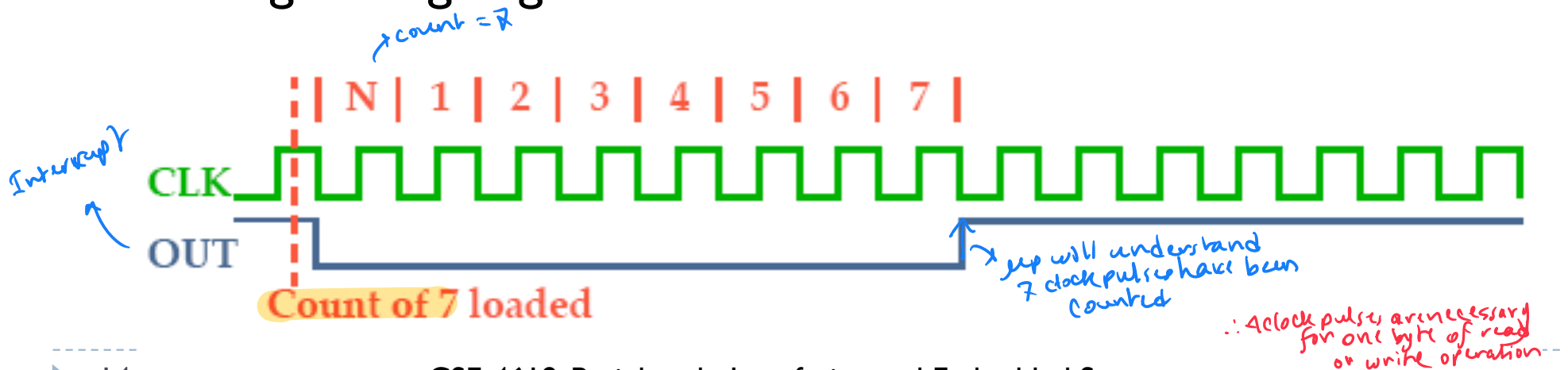
0	<u>Binary Counter 16-bits</u>
1	<u>Binary Coded Decimal (BCD) Counter</u> <u>4 Decades</u>

8254 Modes of Operation

- ▶ Mode 0 (Interrupt on Terminal Count)
- ▶ Mode 1 (Programmable Monoshot)
- ▶ Mode 2 (Rate Generator)
- ▶ Mode 3 (Square Wave Generator)
- ▶ Mode 4 (Software Triggered Strobe)
- ▶ Mode 5 (Hardware Triggered Strobe)

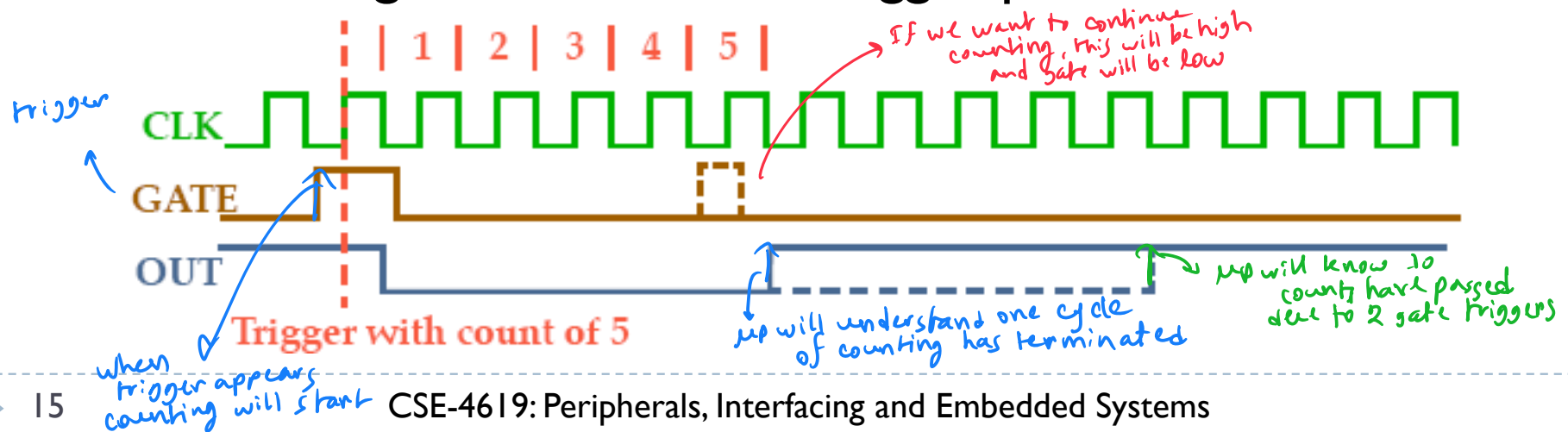
Mode 0: Interrupt on Terminal Count

- ▶ The **OUT** becomes a **logic 0** when the **control word** is written **remains low** even after **count value loaded** in counter.
- ▶ Counter starts decrementing after falling edge of CLOCK
- ▶ The **OUT** goes high upon reaching the **terminal count** & remains high till reloading OUT can be used as interrupt
- ▶ Gate high → normal counting
- ▶ Gate low → counting terminated and current count latched till GATE goes high again



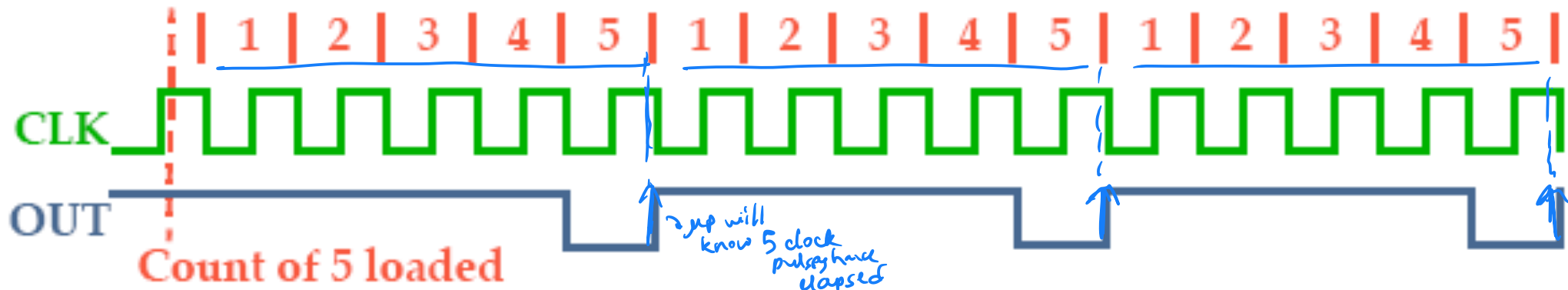
Mode 1: One-Shot Mode *→ can be used for counting continuously*

- ▶ Monostable Multivibrator
- ▶ GATE as HIGH input is used as a trigger input
- ▶ OUT remains high till the count is loaded after application of trigger, OUT goes LOW and remains LOW till count becomes zero
- ▶ Another count loaded, when OUT is already LOW → it does not disturb counting until a new trigger is applied at the gate
- ▶ New counting starts after new trigger pulse



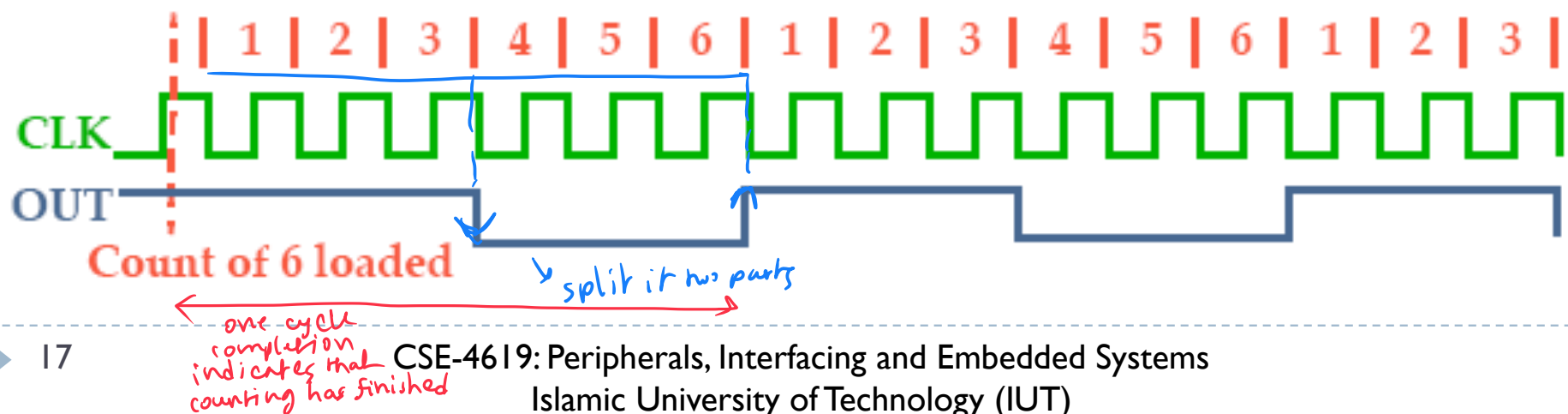
Mode 2: Rate Generator/Divide by N Counter

- ▶ When N is loaded as count → after N pulses → **OUT goes low for only one clock cycle** → then, count N is reloaded → OUT becomes high for N clock pulses
- ▶ The number of clock pulses between the two low pulses is equal to the count loaded
- ▶ Gate → logic 0 → no counting
- ▶ Gate → logic 1 → normal counting



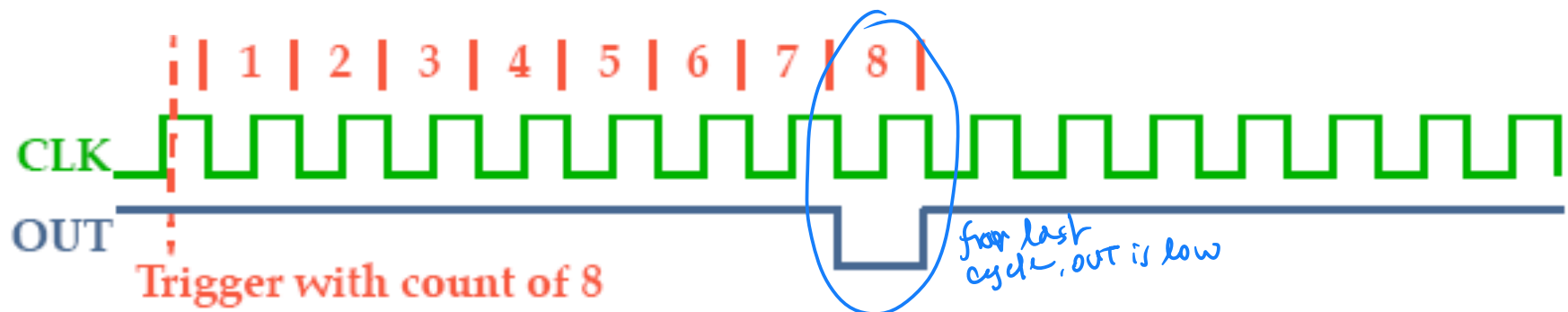
Mode 3: Square Wave Rate Generator

- ▶ When count N loaded is even → output remains HIGH for half the count and LOW for the rest half of the count
- ▶ When count N loaded is odd → output remains HIGH for $(N+1)/2$ and low for $(N-1)/2$.
- ▶ Repeated operation gives square wave
- ▶ Generates a continuous square-wave with G set to 1.
- ▶ If count is even, 50% duty cycle otherwise OUT is high 1 cycle longer.



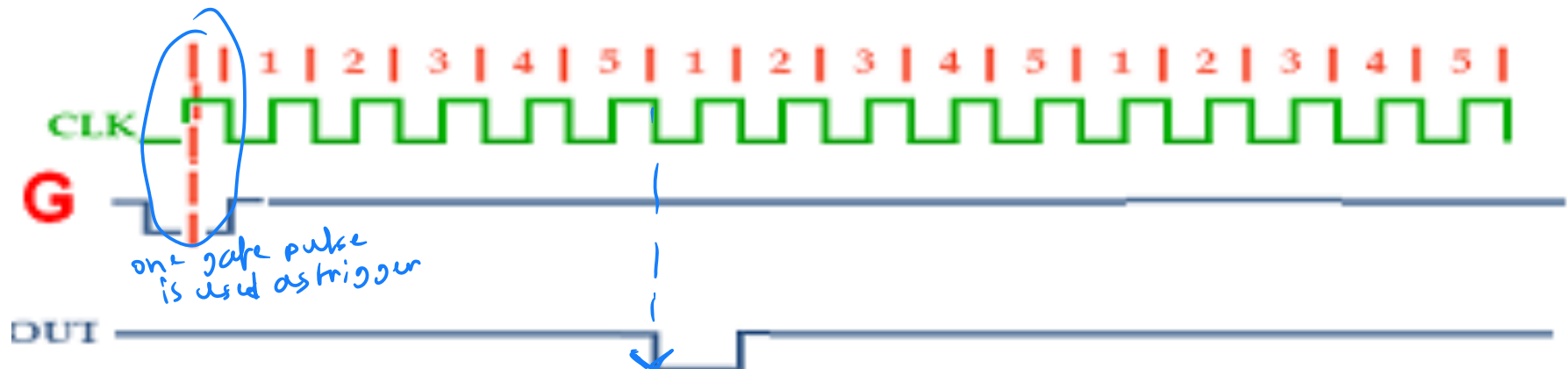
Mode 4: Software Triggered Strobe

- ▶ After mode is set output goes high
- ▶ When count is loaded counting down starts. on reaching terminal count output goes low for only one clock cycle, and then again output goes HIGH
- ▶ The above said low pulse can be used as a strobe for interfacing MP with peripherals
- ▶ When GATE is LOW → counting is inhibited and count is latched. If a new count is loaded while counting, previous counting stops and new counting started in next clock cycle



Mode 5: Hardware Triggered Strobe

- ▶ This mode generates a strobe in response to the rising edge at the trigger
- ▶ Once mode is programmed and counter loaded, OUT goes HIGH
- ▶ Counter starts counting after the rising edge of the trigger (GATE)
- ▶ The OUTPUT goes LOW for one clock period, when the terminal count is reached
- ▶ Output will not go LOW until the counter content becomes zero after the rising edge of any trigger
- ▶ GATE is used as trigger input



Thank You !!

